Byung-cheol Kang

Bioinformatics Research & Development · Genomics · Bacteriophages

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PROFILE

I am a bioinformatics researcher and developer who analyzes genomic and bacteriophage data and builds research pipelines and databases. I use AI coding tools to implement analysis pipelines, develop phage analysis and database tools, and automate repetitive research tasks.

6+ years
Applied bioinformatics

4 profiles
16S · ITS · WGS · Hybrid

98+
Archived project-order lower bound

7 papers
SCI(E) publication record

EXPERIENCE

MicrobiotiX · Bioinformatician — 2026.07 - PRESENT

- Analyze bacterial genomes for host context and phage genomes for quality, annotation, taxonomy, lifestyle, packaging, and tail-gene interpretation.
- Built and evaluated a **Klebsiella pneumoniae** tail/RBP and KL-type resource with 7,759 curated sequences, traceable sources, confidence labels, and verified targets explicitly flagged.
- Apply Phold, ProstT5, Foldseek, and experimental PDB references to structure-informed annotation and engineered-RBP assessment.
- Integrated CheckV, Pharokka, Phold, BACPHLIP, PhaTYP, PhaGCN, and PhageTerm in an anonymized four-genome study that converged on lytic Przondovirus genomes with short DTR packaging.
- Develop tail-RBP design candidates using anchor compatibility, capsule-target evidence, taxonomic context, cluster conflicts, and structural retention.

Wittgenbio · Computational Biology Research — 2026.07.01 - 2026.10.01

- Developed and evaluated bulk-to-single-cell inference approaches that connect bulk expression profiles with curated single-cell references while retaining validation and uncertainty evidence.
- Contributed to virtual-cell research through computational cell-state representation, model evaluation, and biologically grounded interpretation of predicted states or responses.

Sanigen · Bioinformatics Team Lead · Senior Researcher — 2020.03 - 2025.10

- Led a five-person bioinformatics team delivering microbial WGS, metagenomics, assembly, annotation, comparative-genomics, and research support.
- Designed and maintained a shared Snakemake platform covering 16S, ITS, WGS, and Illumina-ONT hybrid assembly with validation, logging, quality gates, and report packaging.
- Standardized execution with Conda, Docker, and Apptainer across Linux compute environments; wrote Python, R, and Bash utilities for QC, analysis, visualization, and reporting.
- Supported comparative-genomics marker discovery and foodborne-pathogen NGS panel development that contributed to two 2023 papers and one joint patent.
- Coordinated analysis review and delivery across a conservative lower bound of 98 archived project orders.

SELECTED PROJECTS

PLATFORM **Integrated microbial genomics platform**
Unified 16S, ITS, WGS, and hybrid profiles under a shared Snakemake execution contract. Active tooling included FastQC, MultiQC, QIIME 2/ DADA2, SPAdes, Flye/Unicycler, QUAST, BUSCO, Prokka, Abricate, eggNOG-mapper, and custom reports.

RESEARCH **Comparative-genomics marker discovery**
Narrowed complete-genome differences to an approximately 17 kb candidate region while separating computational prioritization from downstream assay validation.

PHAGE **Klebsiella phage genomics and tail-RBP design**
Connected host and phage genomics to KL-type evidence, protein structure comparison, and reproducible design exports. Prioritized one candidate with a 129-aa anchor alignment spanning 99.2% of the region and experimental structure support.

WITTGENBIO **Bulk-to-single-cell and virtual-cell research**
Connected bulk expression profiles to curated single-cell references for cell-composition and state inference, and evaluated computational cell models against held-out and biologically interpretable evidence.

TECHNICAL SKILLS

Genomics	Bacterial WGS, phage genomics, assembly, annotation, comparative genomics, KL typing
Phage and protein	Tail/RBP analysis, depolymerase curation, Phold, ProstT5, Foldseek, PDB comparison
Cell-state modeling	Bulk-to-single-cell inference, single-cell reference integration, virtual-cell modeling and evaluation
Workflow engineering	Snakemake, Nextflow, Conda, Docker, Apptainer, Linux
Programming	Python, R, Bash, Git
Selected tools	CheckV, Pharokka, BACPHLIP, PhaTYP, PhaGCN, PhageTerm, Kaptive, BLAST

EDUCATION

M.S., Medical Science · Kyungpook National University School of Medicine · 2016.02 - 2018.08
B.S., Biotechnology; Minor in Psychology · Kyungpook National University · 2008.03 - 2016.02

SELECTED PUBLICATIONS & IP

- Park et al. Novel next generation sequencing panel method for multiple detection and identification of foodborne pathogens in agricultural wastewater. **Frontiers in Microbiology** 14 (2023).
- Park et al. Development and evaluation of a next-generation sequencing panel for pathogens in fermented foods. **Journal of Microbiology and Biotechnology** 33(1) (2023).
- Kang et al. Nothofagin suppresses mast cell-mediated allergic inflammation. **Chemico-Biological Interactions** 298 (2019).
- Joint patent contribution: NGS-based primer set and detection method for foodborne bacteria (2022).

LANGUAGE

Korean - Native English - IELTS 6.5 (October 2023)